

Molecular Element Method (MEM):

A FEM-based formulation for Linear and Non-Linear Molecular Elasticity

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Group of Numerical Methods in Engineering (GMNI)

Tutorial 1: Linear Analysis

<https://andresfernandezsanmiguel.github.io/>



References

-  Fernández-San Miguel, A., Couceiro, I. & Ramírez L. (2026). *The Molecular Element Method (MEM): A FEM-based formulation for Linear and Non-Linear Molecular Elasticity*. University of A Coruña.

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1. Introduction

- What is the Molecular Element Method (MEM)?

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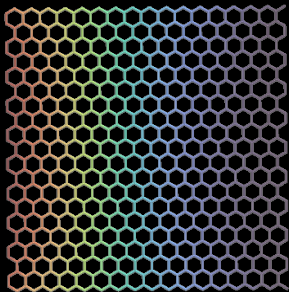
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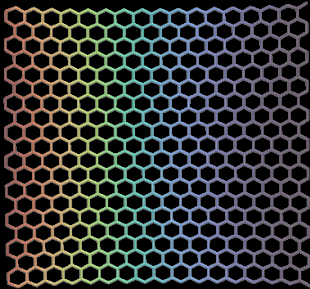
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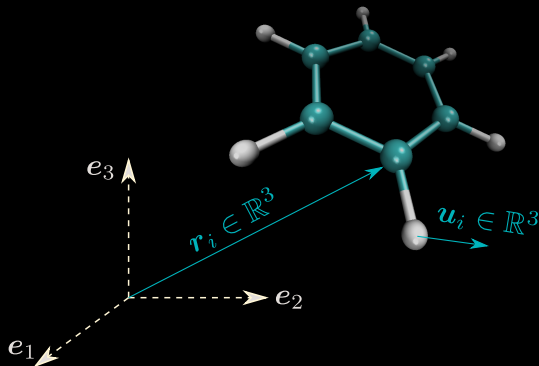
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- Let's consider a molecular structure consisting of N atomic nuclei:

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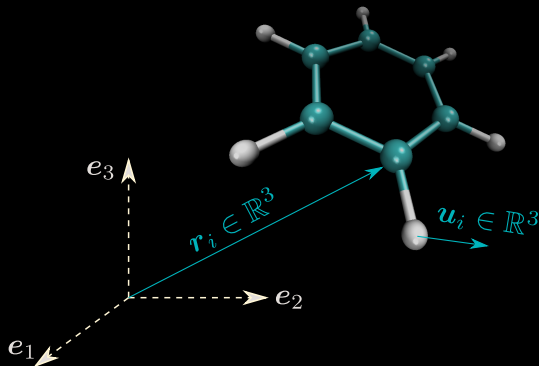
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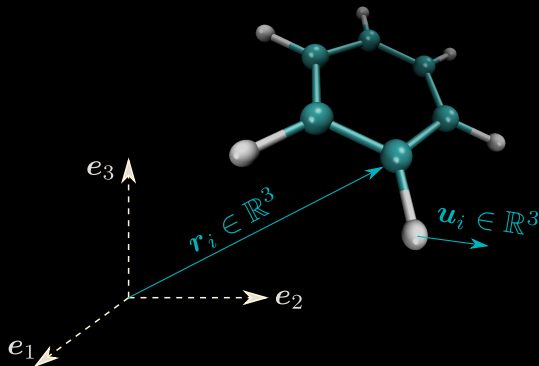
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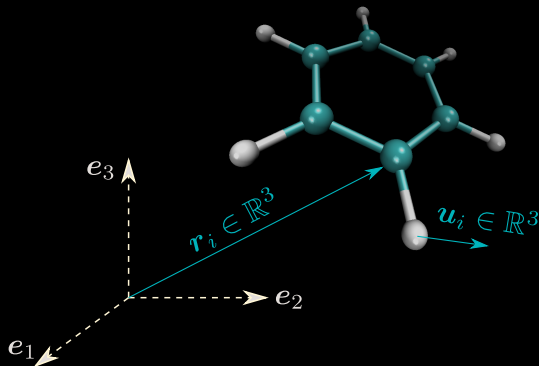
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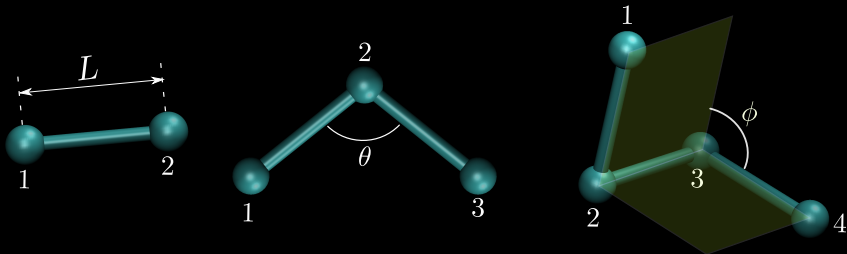
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- ★ Reference configuration $\mathbf{r}^0$, so $\mathbf{r} = \mathbf{r}^0 + \mathbf{u}$

3. Geometrical descriptors

- **Geometrical descriptor:** It is a function φ of a subset of position vectors $\mathbf{r}_\varphi \in \mathbb{R}^{3n_\varphi}$ that describes a geometric measure.

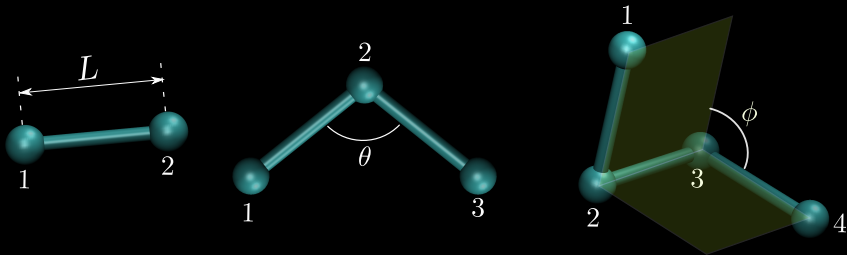
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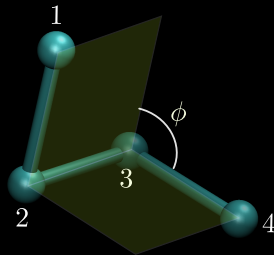
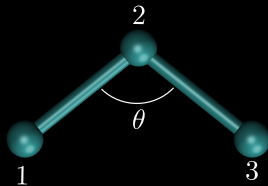
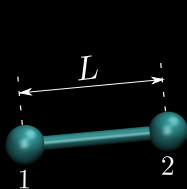
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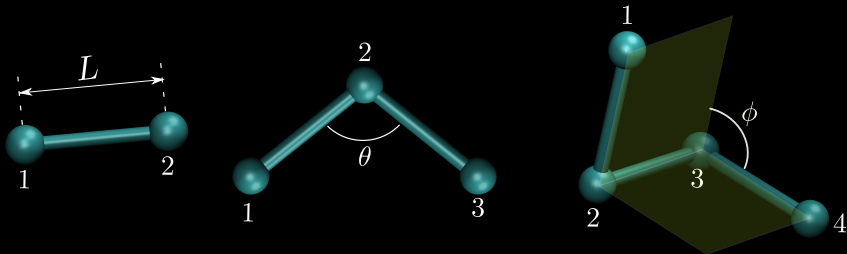
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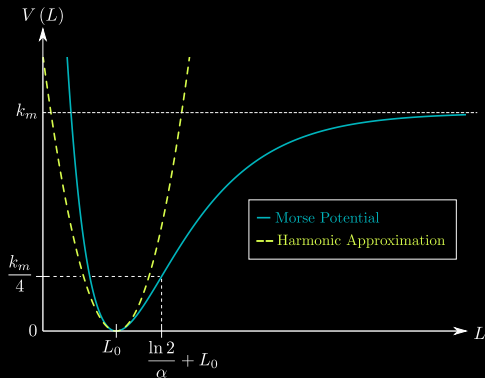
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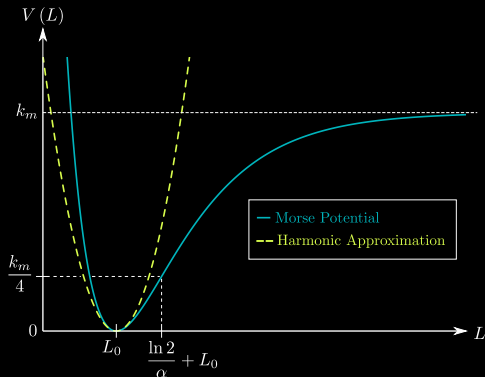


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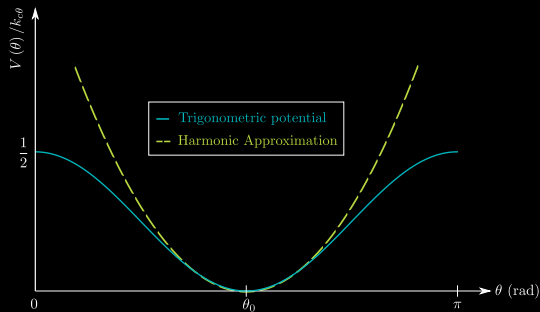
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- Trigonometric Potential $\varphi = \theta$

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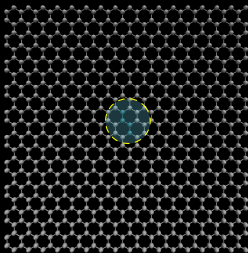
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- ★ $k_\varphi > 0 \forall V_\varphi$ global stiffness matrix Positive Semidefinite.

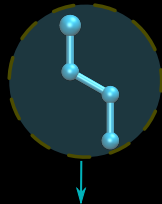
Structure of N atoms



For each potential/element e

$$\left. \frac{d^2 V_{\varphi_e}}{d\varphi_e^2} \nabla_{\mathbf{r}_i} \varphi_e \otimes \nabla_{\mathbf{r}_j} \varphi_e \right|_{\mathbf{r}=\mathbf{r}^0}$$

$\varphi_e \Rightarrow n_{\varphi_e}$ atoms



Local Stiffness matrix

$$\mathbf{K}_{\varphi_e} = \underbrace{\begin{bmatrix} \square & \square & \square & \vdots & \square \\ \square & \square & \square & \vdots & \square \\ \square & \square & \square & \vdots & \square \\ \square & \square & \square & \vdots & \square \\ \square & \square & \square & \vdots & \square \end{bmatrix}}_{\in \mathbb{R}^{3n \times 3n}} \quad \mathbf{K}_{\varphi_e, ij} \in \mathbb{R}^{3 \times 3}$$

Connectivity Matrix

Local Numeration	1	...	i	...	j	...	n_{φ}
Global Numeration	N_1	...	N_i	...	N_j	...	$N_{n_{\varphi}}$

Global Stiffness Matrix

$$\mathbf{K} = \underbrace{\begin{bmatrix} \square & \square & \dots & \vdots & \dots & \square \\ \square & \square & \dots & \vdots & \dots & \square \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \square & \square & \dots & \textcolor{blue}{\square} & \dots & \square \\ \square & \square & \dots & \vdots & \dots & \square \\ \square & \square & \dots & \vdots & \dots & \square \end{bmatrix}}_{\in \mathbb{R}^{3N \times 3N}} \quad \mathbf{K}_{\varphi_e, ij} \in \mathbb{R}^{3 \times 3}$$

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 - ★ **Postprocessing**: Files are generated to visualize the deformation and vibration modes in VMD.

SAOAS
SpectrA

